Project G2: Evaluating Two Decades of NER Progress on CoNLL-2003 (ModernBE CRF)

- **Well-scoped and clearly motivated**: Tracing the progression from dictionary lookup → CRF → BiLSTM-CRF → BERT → ModernBERT on one benchmark is exactly the kind of controlled study that teaches you the most about *why* each architectural generation improved. The ablation framework (document context on/off, CRF head on/off) is clean and well-structured.
- **ModernBERT is a strong choice**: 149M parameters, 8,192 context window, rotary positional embeddings has not been widely benchmarked on CoNLL-2003 yet, so your results will be novel.
- **Clarify the research question now**: You have two independent modifications (document-level context, CRF head). Frame your central question as: *"Do document-level context and structured CRF decoding provide additive or synergistic gains over sentence-level ModernBERT on CoNLL-2003?"* This gives you a clear narrative thread for the final paper.

Recommended actions before Milestone-2 (Most of the materials below will be taught later in the course.)

- **Dataset verification**: Load CoNLL-2003 via HuggingFace `datasets`, confirm train/dev/test sizes (14,987/3,466/3,684 sentences), report entity-type distribution (PER, ORG, LOC, MISC counts), check if `-DOCSTART-` boundaries are preserved in the HuggingFace version—if not, source the original data from the CoNLL-2003 shared task directly.
- **Reproduce published baselines first**: Before any modification, fine-tune `bert-base-cased` on sentence-level CoNLL-2003, confirm you reach F1∼92.4 (within ±0.5 of Devlin et al.). If you cannot reproduce BERT, do not proceed to ModernBERT—debug first.
- **Sentence-level ModernBERT baseline**: Fine-tune `answerdotai/ModernBERT-base` with a standard token classification head (linear layer → softmax), cross-entropy loss, AdamW (lr=2e-5), 5 epochs, batch size=16. Report entity-level F1 (using `seqeval`) on dev and test. This is Experiment 1 in your ablation framework.
- **Implement `-DOCSTART-` concatenation**: Write the document-context loader that concatenates sentences within each article up to 8,192 tokens. Handle the edge case where articles exceed the token limit—implement a sliding window with 128-token overlap. Verify that token-to-label alignment is preserved after concatenation (this is the hardest engineering step—test on 10 sample documents manually).
- **CRF layer implementation**: Use `torchcrf` or implement a CRF layer that replaces the softmax head. Train with negative log-likelihood from the CRF. Verify on a toy example (5 sentences) that transition constraints are learned (e.g., I-PER can only follow B-PER or I-PER, not B-LOC).

Ablations

- 2×2 **factorial design**: You have two factors (document context: on/off; CRF head: on/off), yielding 4 configurations. This is the core of your paper. Report entity-level F1 for all 4, plus per-entity-type F1 (PER, ORG, LOC, MISC) to see which entity types benefit most from each modification.
- **Sliding window parameters**: If using document-level context, compare chunk sizes (512 vs 2048 vs 8192 tokens) and overlap (0 vs 128 tokens)—hypothesis is larger context helps for long-range coreference (e.g., "Obama" mentioned 3 paragraphs earlier resolves "he" as PER), but marginal gains plateau beyond 2048.
- **CRF transition analysis**: After training ModernBERT+CRF, visualize the learned CRF transition matrix as a heatmap. Show that illegal transitions (B-LOC → I-PER) have near-zero probability, and discuss whether the CRF is enforcing constraints the softmax head violates.

Risks & mitigations

- **Risk**: `-DOCSTART-` boundaries removed from HuggingFace version, blocking document-level concatenation—**Mitigation**: Download original CoNLL-2003 data from the shared task website or reconstruct document boundaries using sentence IDs in the dataset metadata; verify on 5 articles that concatenation is correct before running full experiments.
- **Risk**: ModernBERT+CRF gains are negligible (<0.3 F1 over softmax), making the CRF contribution hard to argue—**Mitigation**: This is a valid finding! Frame it as "structured decoding is no longer necessary when pretrained representations are sufficiently expressive." Analyze whether the CRF fixes specific error types (boundary errors vs type errors) even if aggregate F1 is flat.
- **Risk**: Document-level context increases sequence length dramatically, causing OOM on GPU—**Mitigation**: Use gradient checkpointing and mixed precision (fp16). If still OOM, cap at 2048 tokens (not 8192) and note as a limitation. Prototype your chunking logic with a 1,024-token window first to verify correctness before scaling up.
- **Risk**: Integrating a PyTorch CRF layer with a HuggingFace Transformer can be buggy regarding padding tokens—the CRF computes transition scores over the full sequence including `[PAD]`—**Mitigation**: Ensure your CRF loss formulation explicitly masks padding positions. Pass an attention mask to the CRF forward pass so padded positions do not contribute to the partition function or Viterbi decoding.

Open questions

- How will you handle subword tokenization alignment with ModernBERT? ModernBERT likely uses a different tokenizer than BERT's WordPiece—verify that your label-to-token alignment logic works correctly for the first subtoken strategy.
- Will you run each experiment with 3 random seeds and report mean±std (as G19 proposes), or single-seed? Multiple seeds is strongly recommended for a benchmarking study to establish statistical significance.
- The abstract is empty—write it before Milestone-2. A single focused sentence: e.g. "We evaluate whether document-level context and CRF decoding provide complementary gains when added to ModernBERT on CoNLL-2003."
- When using overlapping sliding windows for document context, how will you aggregate the overlapping token predictions during inference? Options include: (a) take predictions from the center of each window (discard edges), (b) average logits across overlapping positions, (c) take the prediction from whichever window places the token furthest from the boundary. Define and justify your strategy.
- In your next report, provide the exact mathematical formulation for the CRF negative log-likelihood loss: $\mathcal{L}_{\text{CRF}} = -\log P(\mathbf{y}|\mathbf{x}) = -\left[\sum_t \phi(y_{t-1}, y_t, \mathbf{x}) - \log Z(\mathbf{x})\right]$. Understanding the partition function Z is essential for debugging.